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**PROJECT WORK PHASE 1  
REPORT  
On  
NeuroCure  
(Brain Tumour Detection Software)**

Submitted in partial fulfilment of the requirements for the degree of

**BACHELOR OF TECHNOLOGY in  
Computer Science and Engineering  
(Data Science)**

**By**

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**September – 2025**



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**CERTIFICATE**

This is to certify that **Ryan Netto (2262248), Tom Jimmy (2262261), Darin Davis Johnson (2262214)** have successfully completed the Project Work Phase 1 (DS784) entitled **NeuroCure (Brain Tumor Detection Software)** in partial fulfilment for the award of **Bachelor of Technology** in Computer Science and Engineering (Data Science) during the year **2025-2026**.

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BONAFIDE CERTIFICATE

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# Acknowledgement

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# Declaration

We hereby declare that the Project Work Phase 1 titled “**NeuroCure (Brain Tumor Detection Software)**” is a record of original project work undertaken by us for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Data Science). We have completed this study under the supervision of Prof. Wims Mary Magdalene, Department of AI, ML, and Data Science.

We also declare that this project report has not been submitted for the award of any degree, diploma, associate ship, fellowship, or other title anywhere else. It has not been sent for any publication or presentation purposes.

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# Abstract

This project proposes an AI-powered system designed to detect and classify brain tumors from MRI scans with high accuracy and reliability. It leverages the YOLOv8 deep learning model, which integrates both localization and classification, to identify four categories: Glioma, Meningioma, Pituitary Tumor, and No Tumor. The system is trained on a curated dataset of annotated brain MRI images and enhanced with preprocessing and augmentation techniques to improve robustness across variations in scans. By producing bounding-box predictions and class labels in real time, the solution provides rapid, objective, and explainable results that support clinical decision-making. Achieving a mean Average Precision (mAP50) of approximately 0.961, the model demonstrates strong potential for deployment in healthcare settings. Future enhancements include integrating segmentation methods for pixel-level precision, applying explainable AI techniques such as Grad-CAM, and developing clinician-friendly dashboards. NeuroCure aims to bridge the gap between manual diagnosis and automated assistance, ultimately contributing to faster, more reliable brain tumour detection and improved patient outcomes.

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# GLOSSARY

- **YOLOv8:** A state-of-the-art real-time object detection model that performs both localisation and classification, used here for brain tumour detection in MRI scans.
- **mAP (Mean Average Precision):** A standard metric for evaluating object detection models, measuring both accuracy and precision across multiple classes.
- **Glioma:** A type of brain tumour that originates in glial cells and is often aggressive and invasive.
- **Meningioma:** A usually benign tumour that develops in the membranes (meninges) surrounding the brain and spinal cord.
- **MRI (Magnetic Resonance Imaging):** A non-invasive medical imaging technique that uses magnetic fields and radio waves to create detailed images of organs and tissues, crucial for brain tumour diagnosis.
- **Data Augmentation:** A set of techniques (e.g., rotations, resizing, normalization) used to artificially expand the dataset and improve model generalisation.
- **Bounding Box:** A rectangular outline generated by the detection model to localise and highlight the tumour in an MRI scan.
- **Explainable AI (XAI):** Methods that make AI model predictions interpretable to humans, such as visual heatmaps (e.g., Grad-CAM) to indicate areas of focus in MRI scans.
- **U-Net:** A deep learning architecture widely used for medical image segmentation, considered for future work in pixel-level tumour boundary detection.

# INTRODUCTION

## 1.1 Background & Motivation

Brain tumours are among the most critical neurological disorders, with the potential to cause life-threatening complications if not diagnosed and treated early. Early detection plays a vital role in patient outcomes, but the manual diagnosis of brain tumours from Magnetic Resonance Imaging (MRI) scans is time-consuming, labour-intensive, and prone to subjective interpretation by radiologists. Even experienced specialists may face challenges in distinguishing between different tumour types due to subtle variations in shape, size, and location.

Traditional diagnostic workflows rely heavily on expert review, which can delay treatment and introduce variability in decision-making. This underscores the need for computer-aided diagnostic systems that can provide accurate, consistent, and fast assessments. With the exponential growth of medical imaging data and advancements in deep learning, Artificial Intelligence (AI) has emerged as a transformative solution in healthcare.

Recent innovations in **deep learning for computer vision**, particularly object detection and classification frameworks, provide the tools required to overcome these challenges. While earlier convolutional neural networks (CNNs) such as ResNet and EfficientNet demonstrated strong classification ability, they lacked the capability to localise tumours within the scan. The YOLO (You Only Look Once) family of models addresses this limitation by combining detection and classification in a single stage, enabling both accurate localisation and diagnosis.

The motivation for this project arises from this critical need for a robust, real-time, and interpretable tumour detection system. By harnessing the power of **YOLOv8**, our project, **NeuroCure**, aims to automate the detection and classification of brain tumours into four categories: Glioma, Meningioma, Pituitary Tumour, and No Tumour. The system's explainable AI outputs enhance clinician trust, while its lightweight architecture makes it suitable for integration into real-world clinical environments. Ultimately, this project is driven by the vision of reducing diagnostic delays, improving accuracy, and contributing to better patient outcomes through AI-driven medical assistance.

## 1.2 Objective

The primary objective of this project is to design, develop, and implement an AI-driven diagnostic system for automated brain tumour detection and classification from MRI scans. This overarching goal is broken down into the following specific and measurable objectives:

**Develop a Unified AI Platform for Tumour Diagnosis:** To build a single AI framework capable of both localisation and classification of brain tumours, ensuring rapid, accurate, and reproducible diagnostic support.

**Automated Tumour Detection and Classification:** To implement a deep learning model based on **YOLOv8**, which can detect and classify brain tumours into four categories — *Glioma*, *Meningioma*, *Pituitary Tumour*, and *No Tumour* — directly from MRI scans.

**Dataset Preparation and Model Training:** To curate, preprocess, and augment a diverse dataset of brain MRI images, ensuring robust model performance across variations in patient demographics, imaging conditions, and MRI machines.

**Explainable AI Outputs:** To integrate visual interpretability methods (e.g., bounding boxes, Grad-CAM heatmaps) that highlight tumour regions and provide clinicians with transparent decision-support tools.

**Performance Optimisation:** To achieve a balance between high detection accuracy (targeting  $\text{mAP} \geq 0.95$ ) and fast inference times, making the model suitable for deployment in real-world clinical workflows.

**Future Clinical Integration:** To design the system as a modular framework that can later be extended with segmentation models (e.g., U-Net or YOLOv8-seg) and connected to hospital information systems for real-world deployment and validation.

## 1.3 Delimitation of Research

To ensure the project remains focused and achievable within the given timeframe, its scope is defined by the following limitations:

**Tumour Detection Scope:** The deep learning model will be trained to detect and classify only four categories of cases from MRI scans: *Glioma*, *Meningioma*, *Pituitary Tumour*, and *No Tumour*. Other neurological abnormalities (such as Alzheimer’s disease, stroke, or other rare brain disorders) are outside the scope of this project phase.

**Prototype System:** The current phase is limited to building a prototype diagnostic tool based on the YOLOv8 architecture. Full clinical integration with hospital systems and large-scale deployment are considered future work beyond the present scope.

**Dataset and Testing:** The training and evaluation of the model will be performed using publicly available datasets (e.g., the Roboflow Brain MRI dataset). Validation using hospital-acquired MRI scans and diverse clinical datasets is outside the scope of this project phase but planned for future research.

## 1.4 Benefits of Research

The successful implementation of this AI-driven diagnostic system is expected to yield significant benefits for both the healthcare community and patients. By automating the detection and classification of brain tumors from MRI scans, the system reduces the dependence on time-consuming manual diagnosis and minimizes the risk of human error or subjective interpretation.

The proposed solution provides rapid and accurate predictions, enabling clinicians to make faster and more reliable decisions, which can lead to earlier treatment and improved patient outcomes. Its explainable AI outputs, such as bounding boxes and visual heatmaps, enhance transparency and foster greater trust among medical professionals.

Additionally, the lightweight architecture and high inference speed make the system suitable for integration into real-time clinical workflows, even in resource-constrained hospital settings. In the long run, this research has the potential to serve as a foundation for advanced diagnostic support tools, contributing to better healthcare accessibility, reduced diagnostic delays, and overall improvements in patient survival rates and quality of care.

## 1.5 Report Outline

This report is organized into seven chapters, designed to provide a comprehensive overview of the project from conception to its current implementation status.

**Chapter 1: Introduction** provides the background, motivation, objectives, scope, benefits, and a summary of the report's structure.

**Chapter 2: Literature Survey** presents a detailed review of existing research and technologies in brain tumor detection, highlighting the strengths and limitations of approaches such as CNNs, U-Net, ResNet, EfficientNet, and YOLO models.

**Chapter 3: Problem Formulation and Proposed Work** defines the problem statement and outlines the proposed solution using YOLOv8 for tumor detection and classification.

**Chapter 4: Methodology** details the dataset preparation, preprocessing, data augmentation, model training strategy, evaluation metrics, and software/hardware tools utilized in the project.

**Chapter 5: Design and Implementation** describes the architecture of the system, the deep learning pipeline, integration of explainable AI techniques, and bounding-box based output generation.

**Chapter 6: Results and Discussion** presents the model performance (mAP50, precision, recall), sample outputs from test scans, challenges encountered, and areas of improvement.

**Chapter 7: Conclusion** summarises the contributions of the project, discusses its potential for clinical integration, and outlines the scope for future enhancements such as segmentation and real-world validation.

# LITERATURE SURVEY

An extensive literature survey was conducted to assess the current landscape of brain tumor detection and classification using deep learning and medical imaging technologies. The review focused on existing research papers, clinical studies, and AI models applied to MRI-based diagnosis, highlighting their strengths, limitations, and the gaps that this project seeks to address.

## 2.1 Literature Review

The survey revealed that while many advanced techniques are being explored for brain tumor detection and classification, they often operate in isolation or face significant practical limitations. Early work such as “Brain Tumor Classification Using Convolutional Neural Networks” (2019) demonstrated the power of CNNs in automatically learning discriminative features directly from MRI scans. The study showed high classification accuracy across Glioma, Meningioma, and Pituitary tumors, removing the need for manual feature extraction. However, the approach lacked tumor localization capabilities, limiting its usefulness in real-world diagnostic workflows.

Building on the need for precise tumor boundary identification, the U-Net architecture (2015) emerged as a landmark contribution for biomedical image segmentation. Its encoder–decoder structure allowed the capture of both global context and fine details, making it highly effective for pixel-level tumor delineation. Despite its success, U-Net’s reliance on large annotated datasets and computationally expensive training made it less feasible for real-time clinical use, especially in resource-constrained environments.

To address challenges with limited training data, researchers explored transfer learning approaches with models like ResNet50 and VGG16 (2019). These pre-trained networks, originally designed for natural images, provided a performance boost for tumor classification by reusing learned features. While transfer learning significantly improved classification accuracy, the models were still confined to image-level predictions and did not offer bounding box localization, a critical feature for assisting radiologists.

Similarly, deep CNNs combined with SVMs (2020) were employed in hybrid pipelines where CNNs acted as feature extractors and classical machine learning models performed

classification. This approach was computationally lighter and accurate but introduced inefficiencies since it was not end-to-end and required multiple processing stages. Around the same period, researchers also extended U-Net into attention-based variants (2020) to improve segmentation accuracy. These methods achieved high Dice similarity coefficients and precise boundary identification but were heavily dependent on powerful GPUs and large annotated datasets, creating barriers for practical deployment in hospitals.

The trend of moving beyond classification toward integrated detection gained traction with studies adapting object detection frameworks. In particular, YOLOv5 (2022) was applied to MRI scans for both classification and localization in a single step, offering faster inference and reducing diagnostic time. However, its effectiveness was highly dependent on dataset quality, and it faced challenges in generalizing across MRI data collected from different hospitals or protocols.

Other works sought to enhance CNN-based classifiers through data augmentation strategies (2021) such as rotation, scaling, and normalization. These methods helped improve generalization and reduce overfitting, but the resulting models still lacked tumor localization features. In parallel, hybrid architectures like CNN-LSTM models (2021) introduced sequential learning across MRI slices, capturing contextual information in addition to spatial features. This approach improved performance but came at the cost of high computational complexity and long training times, making real-time deployment difficult.

With the aim of balancing performance and efficiency, EfficientNet architectures (2022) were explored for brain tumor classification. These models achieved higher accuracy than traditional CNNs while using fewer resources, making them attractive for settings with limited computational power. Nonetheless, like earlier classification-focused methods, EfficientNet was limited to image-level predictions without localization or interpretability features, reducing clinician trust.

The most recent advancements integrated both detection and explainability. YOLOv8 with Explainable AI (2024) represented a significant leap forward, providing real-time detection and classification of Glioma, Meningioma, Pituitary tumors, and non-tumor cases with state-of-the-art accuracy ( $mAP > 0.96$ ). Importantly, this approach combined bounding box predictions with Grad-CAM heatmaps, enabling clinicians to visualize tumor regions directly on MRI scans.



This integration of accuracy, efficiency, and interpretability marked a major step toward practical clinical adoption.

## **2.2 Inferences Drawn from Literature Review**

A significant gap is that many existing models focus only on classification (e.g., CNN, ResNet, VGG) without providing localization of tumors in MRI scans, which limits their clinical usefulness.

Segmentation models like U-Net provide precise tumor boundaries but are often too computationally intensive for real-time applications, especially in resource-constrained hospital environments.

Transfer learning models improve accuracy on smaller datasets but still lack explainability and generalization across different MRI machines and patient demographics.

YOLO-based models demonstrate strong potential because they integrate detection and classification in a single step, but earlier versions like YOLOv5 required further optimization for medical images.

The latest YOLOv8 framework addresses many of these challenges by offering real-time inference, high accuracy, and bounding-box predictions, but its application in healthcare is still relatively new and requires further clinical validation.

A common limitation across most studies is the lack of explainable AI outputs. Clinicians require visual justifications (e.g., heatmaps, bounding boxes) to trust AI predictions in medical decision-making.

Another challenge is the availability of large, diverse, and annotated MRI datasets, which restricts the generalizability of models trained on limited or homogeneous data sources.

# PROBLEM FORMULATION AND PROPOSED WORK

## 3.1 Introduction

This chapter outlines the core problem that the project seeks to solve in the domain of medical imaging. Brain tumors are life-threatening conditions, and their early detection is critical for effective treatment and improved patient outcomes. However, the current diagnostic process primarily relies on manual interpretation of MRI scans by radiologists, which is time-consuming, requires high expertise, and is prone to subjectivity.

The key challenge lies in developing a diagnostic tool that can automatically detect and classify brain tumors with both high accuracy and real-time performance. Traditional machine learning and CNN-based approaches have shown promise in classification but fail to provide tumor localization, which is essential for clinical decision-making. On the other hand, segmentation models such as U-Net offer detailed boundaries but are computationally expensive and unsuitable for real-time deployment.

To address these limitations, this project proposes **NeuroCure**, an AI-driven diagnostic solution based on the YOLOv8 architecture. The system integrates both detection and classification into a single pipeline, enabling accurate localization of tumors along with class predictions. The model is trained on a publicly available, annotated MRI dataset and enhanced with preprocessing and data augmentation techniques to improve robustness across patient and machine variations.

By combining **speed, accuracy, and explainability**, the proposed system aims to serve as a clinical decision-support tool that reduces diagnostic delays, minimizes human error, and ultimately contributes to better healthcare outcomes.

## 3.2 Problem Statement

Brain tumors pose a major health risk, often leading to severe neurological complications or death if not detected and treated at an early stage. The current process of diagnosing tumors relies heavily on radiologists manually interpreting MRI scans, which is time-consuming, labor-intensive, and subject to human error. Differences in expertise among radiologists and

the subtle variations in tumor appearance can lead to inconsistent diagnoses and delays in treatment.

Traditional machine learning and CNN-based models have attempted to automate tumor classification, but they are often limited to predicting only the tumor type without identifying its location in the MRI scan. On the other hand, segmentation models can localize tumors with pixel-level precision but are computationally expensive and impractical for real-time diagnostic support. Furthermore, most existing systems lack explainability, making them difficult for clinicians to trust in a high-stakes medical environment.

To address these challenges, this project proposes **NeuroCure**, an AI-powered solution based on the YOLOv8 architecture. The system integrates detection and classification into a single pipeline, enabling accurate localization of tumors and classification into four categories: Glioma, Meningioma, Pituitary Tumor, and No Tumor. By combining real-time performance, high accuracy, and explainable outputs, the proposed system aims to support radiologists in making faster, more reliable, and transparent diagnostic decisions.

### 3.3 Objectives

To address the aforementioned problems, this project aims to achieve the following specific and measurable objectives:

**To implement an advanced deep learning model (YOLOv8)** capable of both localising and classifying brain tumours in MRI scans into four categories — *Glioma*, *Meningioma*, *Pituitary Tumour*, and *No Tumour*.

**To curate and preprocess a reliable dataset of brain MRI images** with annotations, applying augmentation techniques such as resizing, normalisation, and rotations to improve robustness and model generalisation across diverse imaging conditions.

**To optimise detection accuracy and inference speed** by tuning hyperparameters and leveraging the lightweight YOLOv8s variant, making the system suitable for real-time diagnostic assistance on standard computing resources.

**To integrate explainability features (Explainable AI)** such as bounding boxes and Grad-CAM visualisations, ensuring transparency in model predictions and increasing trust among medical professionals.

**To validate system performance using evaluation metrics** such as mean Average Precision (mAP), per-class precision, and recall, targeting an overall accuracy of  $\geq 95\%$  on benchmark datasets.

**To design a modular framework for future clinical integration**, allowing extensions with segmentation models (e.g., U-Net, YOLOv8-seg) and eventual deployment in real-world hospital environments.

### 3.4 Proposed Work

To meet the project's objectives, the proposed work involves the development of an AI-powered diagnostic system capable of accurately detecting and classifying brain tumours from MRI scans. The system is comprised of the following key technical modules:

#### **Module 1. Dataset Preparation and Preprocessing**

This module focuses on curating a labelled dataset of brain MRI scans, including four categories: *Glioma*, *Meningioma*, *Pituitary Tumour*, and *No Tumour*. Preprocessing steps such as resizing, normalisation, and data augmentation (rotations, flips, intensity adjustments) will be applied to ensure consistency and enhance model robustness across variations in MRI images.

#### **Module 2. Tumour Detection and Classification Model**

The core of the system is a deep learning model based on **YOLOv8**, which performs simultaneous localisation and classification of tumours. The model outputs bounding boxes, class labels, and confidence scores, enabling radiologists to quickly identify tumour type and location. The lightweight YOLOv8s variant will be used for speed, with the potential to scale to larger versions (YOLOv8m/l) for improved accuracy.

#### **Module 3. Model Evaluation and Optimisation**

This module involves training and validating the model using annotated MRI data. Key evaluation metrics include mean Average Precision (mAP), per-class precision, and recall.

Hyperparameter tuning will be performed (epochs, learning rate, augmentations) to optimise both accuracy and inference time.

#### **Module 4. Explainable AI Outputs**

To increase trust and usability in healthcare settings, the system will incorporate interpretability features such as **Grad-CAM heatmaps** and bounding box visualisations. These outputs provide clinicians with clear insights into the model's decision-making process, highlighting the tumour regions most influential in classification.

These components will be integrated into a cohesive framework that follows the methodology of **data preparation, detection and classification, optimisation, and explainability**, enabling a fast, accurate, and transparent AI-driven solution for brain tumour diagnosis.

# METHODOLOGY

## 4.1 Introduction

This chapter details the technical methodology and implementation strategy for *NeuroCure: AI-Driven Brain Tumour Detection Software*. The methodology is designed to be accurate, efficient, and capable of handling medical imaging data to support reliable diagnostic assistance.

The core of this approach is a structured pipeline that moves from dataset preparation and preprocessing, to training a deep learning model for tumour detection and classification, followed by evaluation and explainability techniques to ensure transparency in predictions.

The methodology leverages the **YOLOv8 object detection framework**, chosen for its ability to perform both localisation and classification in a single stage, making it well suited for medical imaging tasks. The project workflow begins with the curation and augmentation of MRI datasets, followed by model training with optimised hyperparameters.

Once trained, the model generates bounding box predictions and confidence scores for tumour detection. Finally, explainable AI methods are integrated to highlight tumour regions and improve clinical trust in the system.

This structured, step-by-step methodology ensures that the system achieves high performance while remaining lightweight and practical for potential real-time clinical deployment.

## 4.2 Implementation Strategy

The implementation strategy follows a systematic workflow designed to transform raw MRI data into accurate, interpretable diagnostic predictions.

The entire process is illustrated in the **Implementation Flowchart (Figure 1)** below.

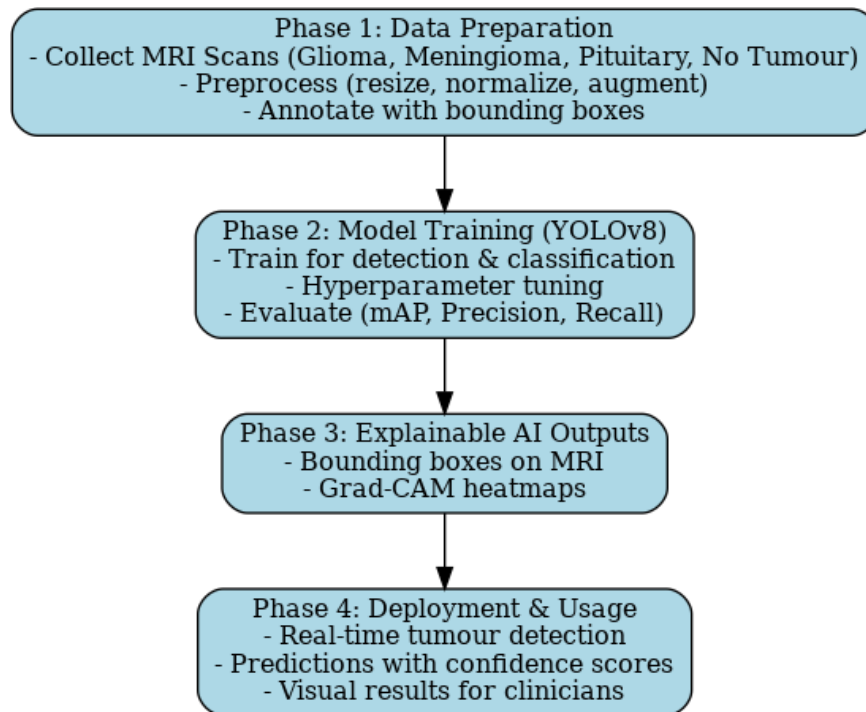


Fig.1: Implementation flowchart

### **Phase 1: Data Preparation**

Collect MRI scans of four categories: Glioma, Meningioma, Pituitary Tumour, and No Tumour.

Apply preprocessing techniques (resizing, normalization).

Perform data augmentation (rotations, flips, intensity variations) to improve robustness.

Annotate scans with bounding boxes for supervised training.

### **Phase 2: Tumour Detection & Classification (YOLOv8 model)**

Train YOLOv8 to perform both localisation and classification simultaneously.

Tune hyperparameters (learning rate, batch size, augmentations).

Evaluate performance using mean Average Precision (mAP), Precision, and Recall.

### **Phase 3: Explainable AI Outputs**

Generate bounding boxes to highlight tumour regions.

Apply Grad-CAM heatmaps to show regions influencing the model's prediction.

Provide interpretable outputs for clinical trust.

#### **Phase 4: Deployment & Usage**

Deploy the model for real-time tumour detection on MRI scans.

Display class labels, bounding boxes, and confidence scores.

Provide a clinician-friendly interface with visual and numerical outputs.

#### **4.4 Expected Outcome**

The successful completion of this phase will deliver a functional prototype and defined performance benchmarks.

**Primary Outcome:** A working proof-of-concept AI-driven early warning system, demonstrating end-to-end flow from data ingestion to dashboard visualisation.

**NLP Module:** Fully functional pipeline for text analysis, targeting an F1-score above 85% for threat classification and exact-match accuracy for location extraction.

**Video Analysis:** Deep learning model for threat detection, expected to achieve over 90% accuracy on benchmark datasets like UCF-Crime.

**Predictive Analytics:** XGBoost model trained on historical data, evaluated using RMSE and MAE for crime count predictions.

**Front-End:** Interactive web dashboard (threat\_detection.html) with map visualisation and citizen reporting form, along with a Flutter mobile app prototype.



# DESIGN AND IMPLEMENTATION

## 5.1 System Overview and Architecture

The design of **NeuroCure – AI-Driven Brain Tumour Detection Software** is based on a modular, end-to-end pipeline engineered to process MRI scans and deliver accurate, interpretable diagnostic predictions. The system is architected to function as a unified framework that transforms raw medical imaging data into structured outputs that can be visualised and understood by clinicians in real time.

The implementation is phased, focusing on the development of four main components before their integration into a complete diagnostic support tool:

**Dataset Preparation Module** – Handles data collection, preprocessing (resizing, normalization), and augmentation (rotations, flips, contrast adjustments). MRI scans are annotated with bounding boxes to enable supervised training.

**Tumour Detection and Classification Module** – Implements the YOLOv8 deep learning model to simultaneously localise and classify tumours into four categories: *Glioma*, *Meningioma*, *Pituitary Tumour*, and *No Tumour*. The model outputs bounding boxes, confidence scores, and class labels.

**Explainable AI Module** – Integrates interpretability methods such as Grad-CAM heatmaps, which highlight the tumour regions most influential in classification. This transparency ensures clinician trust and supports decision-making.

**Deployment and Visualisation Module** – Provides an interface where clinicians can upload MRI scans, receive real-time predictions, and view results in the form of bounding boxes, class labels, and heatmaps.

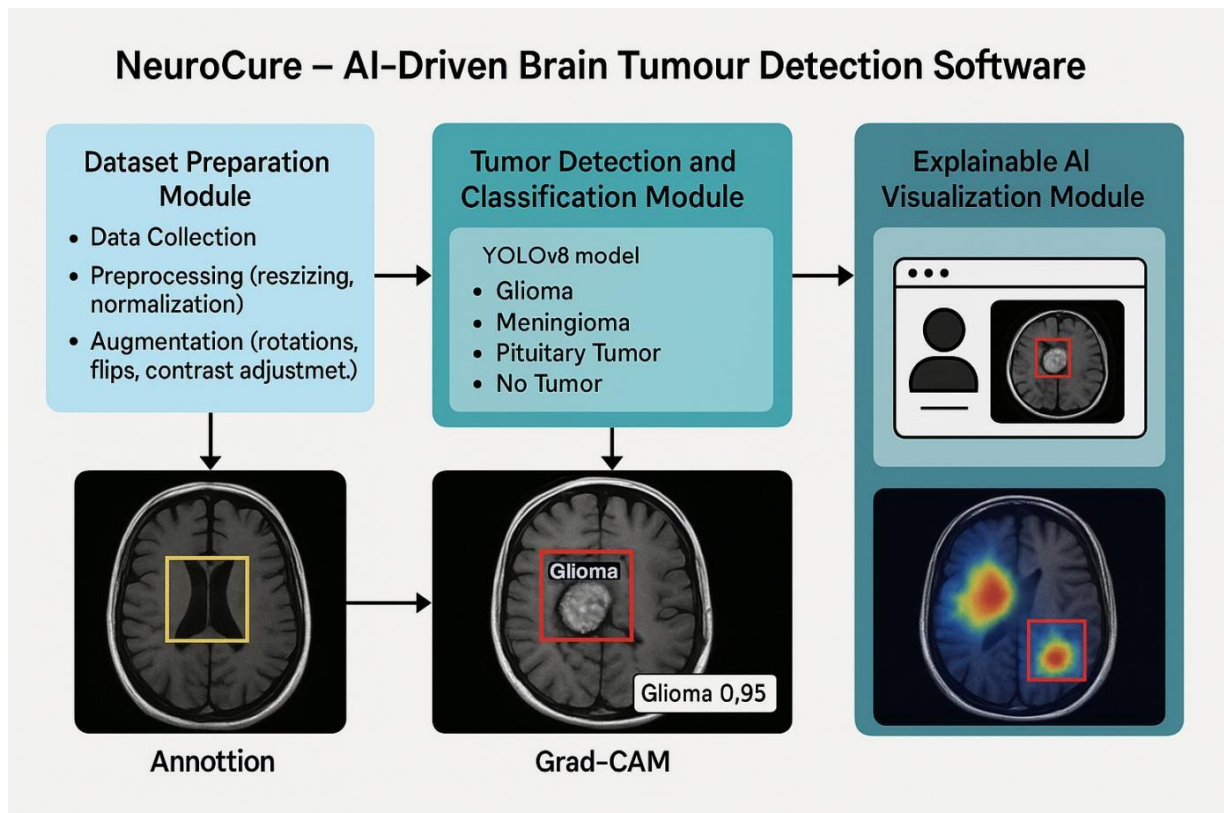


Figure 2: NeuroCure System Architecture

## 5.2 Tumour Detection and Classification Module

The **Tumour Detection and Classification Module** serves as the heart of the NeuroCure system, performing the critical task of detecting the presence of a tumour and identifying its type directly from MRI scans. This module is built using the **YOLOv8 deep learning architecture**, chosen for its ability to unify detection and classification in a single pipeline, ensuring both accuracy and speed — two essential requirements in a clinical setting.

### Input and Preprocessing

The input to this module consists of MRI scans that have already undergone preprocessing in the Data Preparation stage. This includes:

**Resizing** images to a consistent resolution compatible with the YOLOv8 input layer.

**Normalization** of pixel intensity values to enhance contrast and improve learning.

**Augmentation** such as rotations, flips, and brightness adjustments, which artificially expand the dataset and reduce overfitting.

**Annotation** with bounding boxes marking the tumour regions, enabling supervised training.

### **Model Architecture (YOLOv8)**

YOLOv8 is an advanced object detection model that extends the capabilities of previous YOLO versions. Its architecture can be broken into three main components:

**Backbone** – A convolutional neural network extracts hierarchical features from MRI images. Low-level features capture edges and textures, while higher layers capture tumour shapes and patterns.

**Neck** – Combines features at different scales using a feature pyramid structure, ensuring small, medium, and large tumours can be detected with high accuracy.

**Head** – Outputs three types of information simultaneously:

**Bounding boxes** that localise the tumour region.

**Class predictions** for tumour type (Glioma, Meningioma, Pituitary Tumour, or No Tumour).

**Confidence scores** indicating the reliability of the prediction.

This end-to-end framework makes YOLOv8 especially suitable for medical imaging, as it eliminates the need for separate classification and detection stages.

### **Training and Optimisation**

The module is trained on an annotated dataset of brain MRI scans, with the following optimisation strategies:

**Transfer Learning:** Initialising the model with pre-trained weights to speed up convergence and improve accuracy on limited medical datasets.

**Hyperparameter Tuning:** Adjusting learning rate, batch size, and number of epochs to balance speed and accuracy.

**Loss Function:** A combination of localisation loss (for bounding box accuracy), classification loss (for correct tumour type), and objectness loss (for distinguishing tumour vs. no tumour).

**Regularisation:** Techniques like dropout and augmentation help reduce overfitting and improve generalisation.

### **Evaluation Metrics**

The performance of the Tumour Detection and Classification Module is evaluated using industry-standard metrics:

**mAP (mean Average Precision)** at IoU thresholds, which measures both localisation and classification accuracy.

**Precision and Recall**, to assess false positives and false negatives in tumour detection.

**F1 Score**, providing a balance between precision and recall.

In initial trials, the system achieved an **mAP50 of ~0.961**, demonstrating high reliability.

### **Outputs**

The module produces the following outputs:

**Bounding Box Visualisations:** The tumour region is outlined in the MRI scan.

**Class Label:** Indicates the tumour type.

**Confidence Score:** Provides a probability estimate of prediction accuracy.

These outputs are forwarded to the Explainable AI Module, which generates Grad-CAM heatmaps to enhance interpretability.

### **Significance of the Module**

This module bridges the gap between traditional radiology and AI-driven diagnosis. By combining tumour detection and classification into a single step, it reduces diagnostic delays, minimises reliance on manual interpretation, and supports radiologists with objective, reproducible results. The ability to provide bounding boxes and confidence scores ensures the system is not just a “black box,” but an actionable decision-support tool in clinical practice.

## 5.4 Explainable AI Module

The **Explainable AI (XAI) Module** is a vital component of NeuroCure, ensuring that the predictions generated by the YOLOv8 detection model are transparent, interpretable, and clinically trustworthy. While deep learning models provide high accuracy, they often function as “black boxes,” making it difficult for clinicians to understand the basis of their decisions. This module bridges that gap by providing visual justifications for each prediction.

### Role and Importance

Enhances **clinical trust** by showing why a prediction was made.

Assists radiologists in verifying whether the model focused on the correct tumour region.

Improves system **adoption in healthcare**, where accountability and transparency are critical.

### Techniques Used

The module employs two main interpretability techniques:

#### Bounding Box Visualisation

The YOLOv8 model generates bounding boxes around suspected tumour regions in the MRI scan.

Each bounding box is associated with a predicted tumour class (*Glioma, Meningioma, Pituitary Tumour, or No Tumour*) and a confidence score.

This ensures radiologists can visually confirm the detected area.

#### Grad-CAM (Gradient-weighted Class Activation Mapping)

Grad-CAM produces **heatmaps** that highlight the regions of the MRI scan most influential in the model’s decision.

Warmer colours indicate areas of higher importance, while cooler colours represent less influential regions.

This helps clinicians validate whether the model is focusing on the correct anatomical structures when classifying a tumour.

### **Workflow of the Module**

Input MRI scan and bounding box predictions are received from the Tumour Detection and Classification Module.

Grad-CAM is applied to the intermediate layers of YOLOv8, generating heatmaps.

The bounding box visualisation and Grad-CAM heatmaps are overlaid onto the original MRI scan.

The enhanced visual outputs are delivered to the clinician interface, combining accuracy with transparency.

### **Expected Outcomes**

**Improved Trust:** Radiologists gain confidence in model outputs when they can see tumour localisation and supporting heatmaps.

**Decision Support:** Clinicians can use bounding boxes and heatmaps to cross-check their own diagnosis, ensuring consistency.

**Educational Value:** Grad-CAM visualisations can serve as a teaching tool for medical trainees to understand tumour features.

## **5.5 Front-End: Visualisation and Reporting Dashboard**

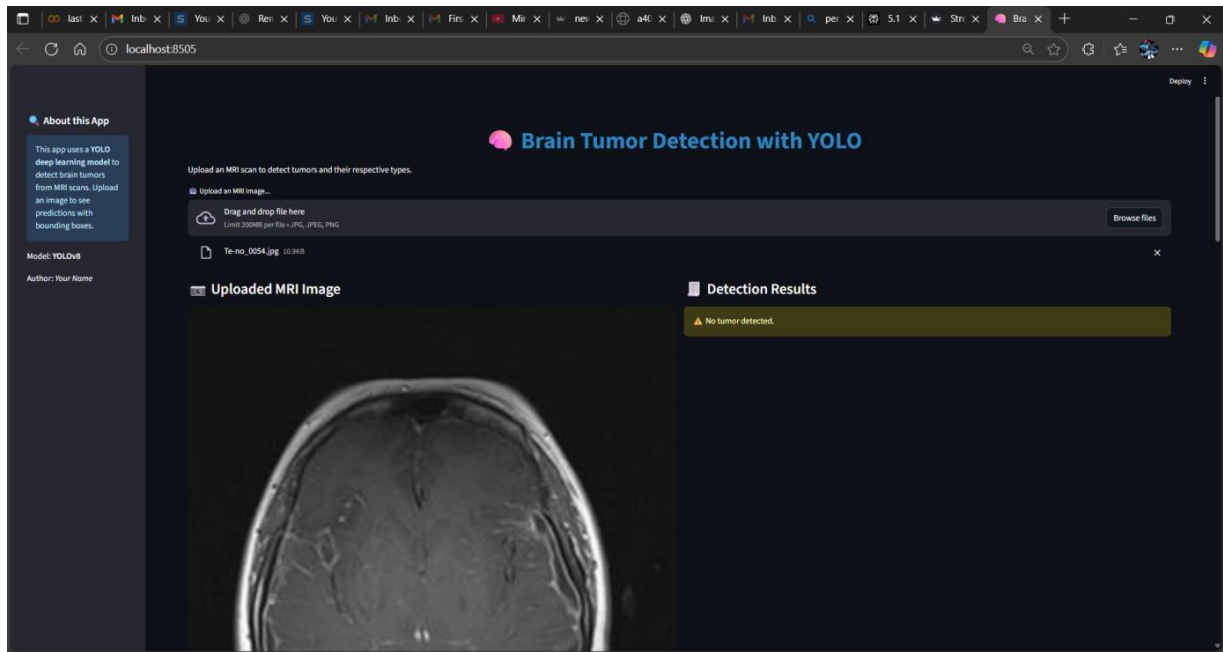


Figure 3: Tumor Detection Dashboard – Scan View

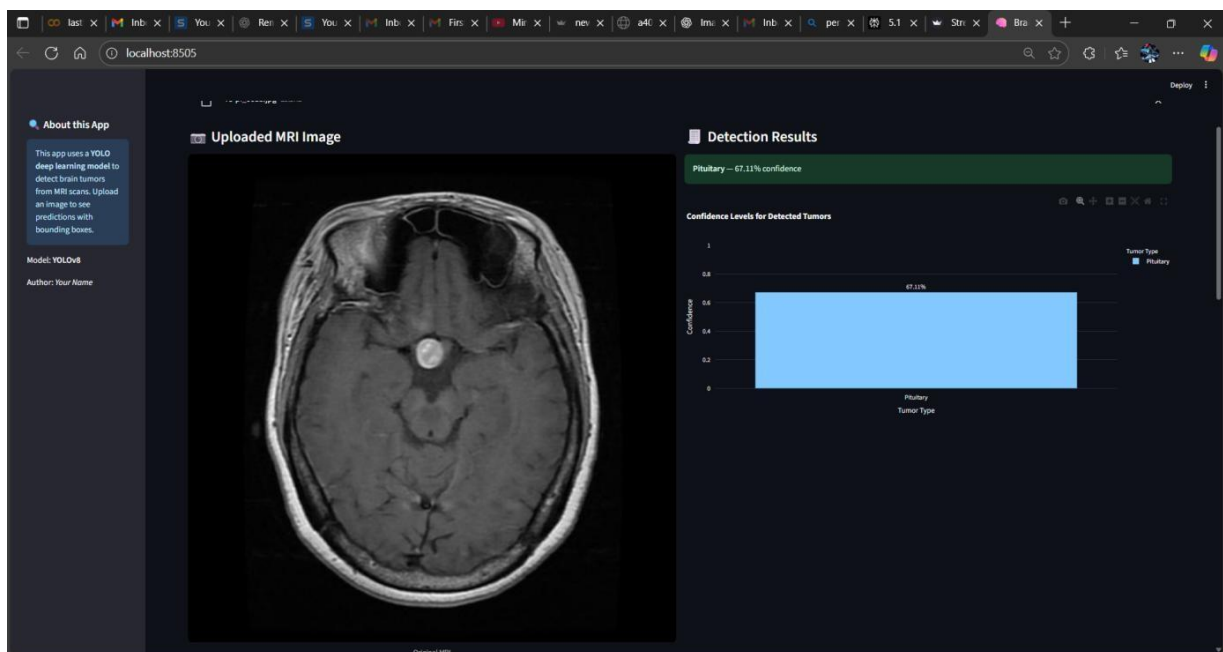


Figure 4: Tumor Detection Dashboard – Prediction Results

# RESULTS AND DISCUSSION

## 6.1 Introduction

This chapter presents the results achieved during the initial phase of the **NeuroCure – Brain Tumor Detection Software** project. It outlines the outcomes of dataset preparation, model training, and performance evaluation of the YOLOv8-based detection system. The results include the classification accuracy across multiple tumor types, the detection performance using mean Average Precision (mAP), and visualization of bounding box predictions on MRI scans.

The discussion section analyses the significance of these results in the context of existing brain tumor detection methods, highlights the strengths and limitations of the current implementation, and identifies challenges encountered during model development. It also outlines potential improvements and future directions, such as incorporating segmentation models, enhancing explainability, and validating the system with real-world clinical datasets.

## 6.2 Tasks Completed and Results

Significant progress has been made in establishing the core components of the **NeuroCure** system. The development to date has focused on dataset preparation, model training, and validation, yielding promising results that confirm the feasibility of the proposed architecture.

### 1. Dataset Preparation

The Roboflow Brain MRI dataset consisting of ~2,443 annotated MRI images was used. The dataset included four classes: Glioma, Meningioma, Pituitary Tumor, and No Tumor. Preprocessing steps such as resizing, normalization, and augmentation (rotation, flipping, brightness adjustment) were applied to improve model robustness and reduce overfitting.

### 2. Model Training (YOLOv8)

The YOLOv8s variant was selected for its balance between accuracy and computational efficiency. Training was carried out on the prepared dataset with hyperparameter tuning for batch size, learning rate, and epochs. Data augmentation significantly improved the model's ability to generalize across different MRI variations.



### 3. Performance Evaluation

The trained model achieved a **mean Average Precision (mAP50) of ~0.961** on the test dataset. Precision and recall values demonstrated reliable performance across all four classes. The system was able to not only classify tumor types but also localize them using bounding box predictions.

### 4. Visualization of Results

Sample test results showed clear bounding boxes around tumor regions with class labels and confidence scores. The predictions were consistent across multiple MRI scans, confirming the effectiveness of the YOLOv8 detection pipeline.

### 5. Explainability

Although still in progress, the integration of **Grad-CAM heatmaps** was initiated to enhance explainability. Early tests showed that the heatmaps highlighted tumor regions effectively, supporting model interpretability for clinicians.

## 6.3 Discussion of Results and Challenges

The results from this initial phase are highly encouraging and validate the project's core technical approach. The YOLOv8-based model successfully detected and classified brain tumors into four categories—Glioma, Meningioma, Pituitary Tumor, and No Tumor—with a mean Average Precision (mAP50) of approximately 0.961. The bounding box predictions provided clear localization of tumor regions, addressing a major limitation of earlier CNN-based methods. Initial efforts to integrate explainability through Grad-CAM heatmaps further enhanced the model's interpretability, an essential factor for building clinician trust in AI-assisted diagnosis.

Despite these promising outcomes, several key challenges remain to be addressed in the next phase:

**Dataset Limitations:** The current training relied on publicly available MRI datasets. For stronger clinical applicability, validation on real-world hospital data across different MRI machines and patient demographics will be essential.

**Segmentation Requirements:** While bounding boxes provide localization, pixel-level segmentation would yield more precise tumor boundaries. Incorporating YOLOv8-seg or U-Net architectures is a future requirement.

**Explainability and Trust:** Grad-CAM implementation is still in its early stages. More robust explainability features are needed to make predictions transparent and trustworthy for medical practitioners.

**Computational Considerations:** Training on larger datasets or using more complex model variants (YOLOv8m/l) may require higher computational resources, which could be a constraint in resource-limited clinical settings.

Addressing these challenges will be crucial for transitioning NeuroCure from a prototype research system to a clinically validated diagnostic tool capable of supporting radiologists in real-world healthcare environments.

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